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On the Aerodynamic Performance of a Blended-Wing-Body, Low-Mach Number Unmanned Aerial Vehicle

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Abstract: A study on aerodynamic design studies of a blended wing–body (BWB) unmanned aerial vehicle (UAV) operating at low Mach numbers is presented. First, a parametric investigation based on analytical equations is carried out to identify the range of the necessary wetted area for the UAV to maximize endurance at a Mach number close to 0.1. A base-of-reference configuration is designed, and its aerodynamic performance is evaluated by utilizing a panel method in Xflr5. An optimization algorithm is then incorporated to trim the UAV and produce the 'clean' configuration. Computational fluid dynamics (CFD) simulations are performed within the OpenFoam environment to produce first the updated drag polars, and then, to analyze the integration of the nacelle and the pair of electric ducted fans (EDFs) used for the propulsion system. In particular, when examining the integration of the nacelle with a spinning electric ducted fan (EDF) standing as the propulsion system of the vehicle, a rotating, sliding mesh computational approach is adopted. Results indicate that the clean configuration is characterized by strong longitudinal stability so that the UAV has the potential to fly trimmed at very low speeds. Mounting EDFs on the back of the fuselage is conducive to higher loading with minimal drag penalty. An increased lift-to-drag ratio is achieved. Reduced wake mixing due to the EDF's jet flow is observed. The spanwise flow that is conducive to pitch brake and loss of stability is also weak, as the suction produced by the EDF diverts the flow inboard.

Keywords: blended wing–body (BWB); unmanned aerial vehicle (UAV); aerodynamic performance; ducted fan propulsion system

Academic Editor: Pier Marzocca

Received: 24 January 2025

Revised: 14 February 2025

Accepted: 16 February 2025

Published: 20 February 2025

Citation: Lampropoulos, N.; Vouros, A.; Templalexis, I.; Lekas, T. On the Aerodynamic Performance of a Blended-Wing-Body, Low-Mach Number Unmanned Aerial Vehicle. *Fluids* **2025**, *10*, 54. https://doi.org/10.3390/fluids10030054

1. Introduction

Thanks to their enhanced aerodynamic efficiency, blended wing–body (BWB) aircrafts are considered the most promising, revolutionary replacements for traditional tube-and-wing aircrafts [1]. Extensive simulations and targeted experiments have been used for the adjustment and refinement of BWB designs of different sizes [2]. BWBs benefit from reduced wetted area and interference drag, with lift generated near the main body, which itself serves as a lifting surface. Moreover, when combined with hybrid or fully electric propulsion systems, they align well with fuel economy and environmental goals. However, challenges related to the integration of engines with the 'clean' aircraft configuration remain. In the review studies by Chen et al. [2], Ordoukhanian and Madni [3] and Okonkwo and Smith [4], those challenges, along with technologies that could potentially improve BWB performance, are discussed. As the greening of air transport becomes more and more significant for future aviation, recent studies have explored the potential of hydrogen-powered blended wing–body aircraft to reduce CO_2 emissions and enhance fuel efficiency through advanced technologies like boundary layer ingestion and ultra-high bypass ratio engines. While findings highlight significant environmental benefits, obstacles related to high operational costs and hydrogen storage limitations must be addressed for practical implementation in sustainable aviation [5].

Liebeck's multidisciplinary optimization framework addressed key challenges in large-scale BWB aircraft design, emphasizing the importance of coupling aerodynamic, structural and control considerations [6]. Subsequent research efforts focused on aerodynamic optimization of various-sized aircrafts. Some of the most important ones include those of Qin et al. [7], Peigin and Epstein [8], Lyu and Martins [9], Kuntawala et al. [10] and Staub et al. [11], who explored configurations to improve aerodynamic efficiency and lift-to-drag ratios and minimize drag through optimization techniques and trade-off analyses.

Spreading BWB design to unmanned aerial vehicles (UAVs) offer the opportunity for cost-effective, energy-efficient multipurpose platforms and maximizing range and endurance, with respect to usage, which can be related to tactical and emergency missions, including border surveillance but also medicine or other cargo delivery to topographically challenging locations [12,13]. The low-speed longitudinal aerodynamic performance of vehicles with unconventional BWB shapes has also been considered in low-Mach spaceships, used for future space stations [14,15]. The Hyperion project, a flagship collaboration between the Universities of Colorado, Sydney and Stuttgart, offered a platform for exploring innovative design ideas and evaluating hybrid electric propulsion systems [16,17]. A research team from the Aristotle University of Thessaloniki analyzed several design aspects of the specific configuration [12,18–20]. Recently, Kapsalis et al. [18] used computational fluid dynamics (CFD) along with a design of experiments (DOE) method to determine the optimal values for aspect ratio, taper ratio and sweep angle, achieving a balance between maximum top speed, minimal takeoff runway requirements and gross takeoff weight. In cooperation with Cranfield University, an institution actively involved in UAV, BWB conceptual design from Ecuador [21–23] presents, apart from electrically driven propulsion concepts, a discretized semi-empirical performance method to analyze the boundary layer ingestion (BLI) as an effect of the coupling of aircraft's aerodynamic configuration with its propulsion system and to discuss fan deterioration due to flow distortion at the inlet.

Considering engine integration and application of the BLI technique, a literature survey shows that both those aspects have been the subject of research of several studies dealing with various types of aircraft, although barely applicable to UAVs [24–28]. In any case, analyzing aerodynamics in parallel to the propulsion system is a very challenging task. There are several methods to analyze effects, which can be classified according, for example, to the degree of aero-propulsive coupling, or the level of modelling, based on the mechanisms that are represented in the set of equations solved [24]. Liou et al. [29] included in the appendix comparisons of results on engine integration produced by the most frequently used approaches such as the actuator disk and the body-force models that was applied to steady-state Reynolds-averaged Navier–Stokes (RANS) simulations. Tse and Hall [30,31] examined the impact of the turbulent boundary layer over the fuselage on the fan inlet based on results of unsteady RANS (URANS). Zhang and Barakos [32,33] demonstrated that in propulsion systems of multiple propellers or electrically distributed electrically powered ducted fans (EDFs), simulations that account for both rotational speed and boundary layer development over the wing can provide the most detailed information about axial flow and pressure distribution as well as thrust production as a function of the advance ratio. Considering in particular the utilization of EDFs, Urban et al. [34] discussed the impact of design modifications related to geometry and fan arrangement on the performance of EDFs for potential use in unmanned aerial vehicles. Extended experimental testing indicated that geometrical adjustments can enhance thrust, reinforcing EDFs as a viable and efficient alternative to conventional propulsion systems.

In the present work, a series of aerodynamic studies of a BWB UAV designed to operate at low Mach numbers is presented. A preliminary, parametric investigation is conducted to identify an initial configuration suitable for the UAV to operate at Mach numbers close to a value of 0.1. Next, a base-of-reference UAV model is built using a series of typical NACA airfoils within the environments of an open-source aerodynamic analysis tool. A three-dimensional (3D) panel method (PM) is used for the estimation of the aerodynamic performance of the model and the evaluation of the analytical calculations of the parametric analysis. An optimization algorithm is then incorporated to refine the base-of-reference configuration. The central body airfoils are reflexed to improve the lift-to-drag ratio and to enhance trimming capabilities at relatively high positive angles of attack. The aerodynamic performance of the optimized, 'clean' configuration is assessed using an incompressible computational fluid dynamics (CFD) solver available in OpenFOAM (v.2412). Two additional computational models are used for studying the integration of the nacelle and the integration of the nacelle together with the ducted fans that stand for the propulsion system. For this latter analysis, the adoption of the specific means of propulsion offers the opportunity to utilize an advanced computational approach, which is free from additional modelling, based on a moving, sliding mesh that accounts for the rotating fan combined with a static mesh used for resolving the flow over the UAV configuration.

2. Aerodynamic Design Concepts and Models

2.1. Design Considerations and Method

The aerodynamic design studies analyzed in the following are part of a proof-of-concept for the development of a BWB UAV operating at low velocities and more specifically in the vicinity of a Mach number close to 0.1. The objective is to provide an initial configuration and examine key concepts and technologies, including, on the one hand, the aerodynamic benefits of the 'clean', i.e., without engine integration, BWB configuration, and on the other, the enhancement of its aerodynamic performance when a pair of ducted fans are incorporated for propulsion.

The aerodynamic features of the clean BWB aircraft have been highlighted in several studies reviewed in the previous section. The central part of the aircraft is a lifting surface by itself. In terms of drag reduction, the interference drag that is generated by the mixing of airflow around the airframe components in a conventional aircraft design is avoided. Moreover, less skin friction drag due to wetted area reduction for the same lift is expected, while the airframe volume is increased compared to that of typical tube-and-wing aircrafts. Due to the delta wing presence, the aircraft is expected to be less prone to air disturbances than conventional ones. As regards stability, although tailless, the specific design exhibits high longitudinal stability due to the aft positioning of the center of pressure with respect to the center of gravity. Reflexing airfoils also facilitate trimming at larger angles of attack, while the adoption of anhedral reduces high-rolling stability due to the swept back wings, which may lead to dutch roll dynamic instability. In terms of mission profile, the produced UAV is expected to operate at low cruise flight speeds while being able to fly near stall speed at high lift flight conditions. To this end, a polyhedral wing with high anhedral could reduce the wingtip-induced drag, which is higher at low speeds, while it reduces the spanwise flow, which leads to pitch-braking for back-swept wings at high angles of attack. Finally, considering integration with the propulsion system, mounting engines to the upper surface of the aircraft results in significantly reduced infrared and acoustic signature levels while, as a next step, the BLI has the potential to enhance aerodynamic and propulsive efficiency.

The above considerations are kept in mind in all the steps of a sequence of aerodynamic studies discussed in the present work. Each step incorporates different methods and tools, starting from analytical equations based on fundamental theories and concluding with advanced computational strategies, as will be analyzed in the following paragraphs. The study presents the results of a series of aerodynamic studies of a BWB UAV designed to operate at low Mach numbers, with a preliminary parametric analysis to identify an initial configuration suitable for Mach 0.1 operation. The study uses a 3D panel method for aerodynamic performance estimation, followed by optimization to refine the configuration, including reflexed central body airfoils for improved lift-to-drag ratio and trimming capabilities. The optimized configuration is assessed using OpenFOAM for CFD analysis, with additional models evaluating the integration of nacelle and ducted fans, utilizing an advanced computational approach to simulate the rotating fan and UAV flow interactions. The progress of the work for the implementation of the above tasks is presented in Figure 1.

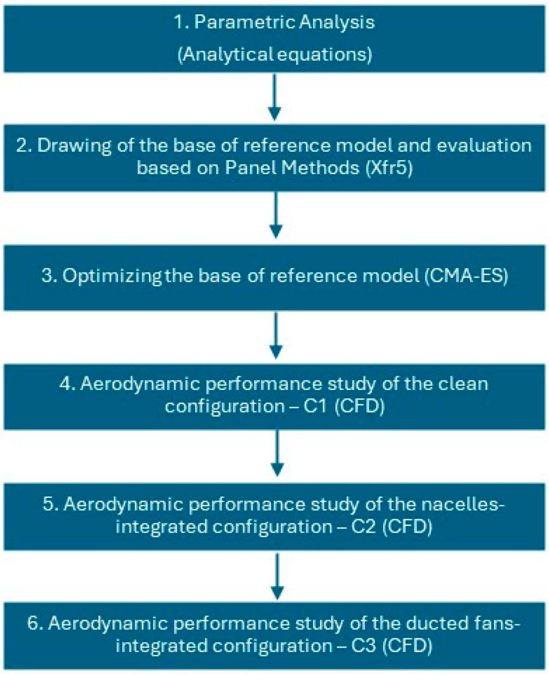


Figure 1. Tasks and analytical/computational tools used in the present work.

2.2. Preliminary Parametric Analysis

Aimed at designing a UAV with the above-discussed characteristics, a parametric investigation was conducted first to identify the range of values of essential design variables which result in a stably operating aircraft at a Mach number of about 0.1. For this purpose, simplified equations coming from the earlier literature and experience were utilized. The main variables of the study are wetted area and flight speed range. By applying conservative approximations to critical aerodynamic properties, this preliminary analysis was used for calculating the maximum lift-to-drag ratios that corresponds to an optimal range of values, with respect to endurance, or range, aircraft. The following steps are considered:

- Regarding the decision on the range of flight speed and the wetted area values, with respect to flight speed, velocity (IAS) values vary between 20 and 120 m/s. For wetted area, values ranging from 5 to 20 m^2 are considered.
- The estimation of the skin friction drag coefficient is based on typical turbulent flat-plate boundary layer flow geometry, i.e., $C_f = 0.074/Re^{1/5}$ [35]. For the Reynolds

number, the mean aerodynamic chord (MOC) of the wetted area is calculated based on the aspect ratio value for BWB configurations (AR = 2). The flat-plate wing equivalent concept is then adopted, for which the skin friction is multiplied by a factor of 2.2 [36].

- The parasitic drag coefficient is calculated assuming that it is 25% higher than the skin friction drag coefficient [32]. The total drag coefficient is the sum of the skin friction and the parasitic drag coefficients.
- The lift coefficient is estimated using the aircraft's weight (205 kg) and the varying flight speeds and planform areas.
- The lift-to-drag ratio (C_L/C_D) is plotted as a function of flight speed for various wetted areas within the range considered. For optimizing endurance, the flight speed that corresponds to $(C_L/C_D)_{max}$ is targeted. $({C_L}^{1/2}/C_D)_{max}$ is, in addition, used in the same way to identify the optimal flight speed that maximizes range.

2.3. Xflr5 Model

Xfr5 (version 6.60) is an open-source aerodynamic and stability analysis tool [37]. It offers the capability to draw aerodynamic models consisting of typical or customized airfoils. At this phase of research, Xflr5 was used for designing the base-of-reference configuration. The model was built 'from scratch' using available NACA airfoils: NACA2412 airfoils were adopted for the central sections and NACA0012 airfoils for the outer ones. Efforts have been made to design a model able of achieving the requirements set by the parametric analysis performed in the previous step. More specifically, the planform area, the aspect ratio and the mean aerodynamic chord were continuously monitored and compared to those that, according to the preliminary analysis, provide optimal endurance (and range) for the range of operating Mach numbers. It has to be noted that Xflr5 was incorporated as a preliminary aerodynamic analysis tool, and it was used for the development of the base-of-reference model only. All the models developed after the optimization process, which are discussed in the following paragraphs, are evaluated with respect to their aerodynamic characteristics based on the results of CFD modelling.

In the present study, Xflr5 was used for the calculation of the lift, drag and moment coefficients of the base-of-reference configuration. For this purpose, a three-dimensional (3D) panel method (PM) was incorporated. This method stands as a refinement of the earlier lifting line theory (LLT) and vortex lattice method (VLM), taking into account the wings' thickness instead of their mean chord [37]. The theoretical background and the details of the computational implementation of the method are described in [38].

2.4. Shape Optimization Algorithm

Aiming at optimizing the airfoils used for the base-of-reference configuration, the covariance matrix adaptation evolution strategy (CMA-ES), as proposed by Hansen et al. [39], was utilized. The algorithm works for the identification of design variable values which vary within a user-defined range that minimize a cost function (see Equation (1)).

$$\text{Cost function} = |\text{MTOW} - \text{Lift}| + |C_D/C_L| + |C_m| \quad (1)$$

The cost function is minimized if the produced lift equals the predefined MTOW, and meanwhile the C_D/C_L is minimized (conditions for maximum endurance) and the aircraft model is trimmed (condition for level flight). The lift, drag and pitching moment about the center of gravity under cruise conditions within a predefined static margin are estimated in a modular manner by making use of the two-dimensional lift-to-drag polars of the candidate airfoils used in the UAV model. The loading of each segment is estimated by a division in ten sub-segments, which encounter the flow with a geometrically calculated constant angle of attack (α). Upon the definition of the local angle of attack for each sub-part, its loading is estimated in a straightforward manner through the polar database and the assumption of lumped loading on the aerodynamic center of the segment sub-part (one quarter of a chord). The design variables are the incidence angle of the root airfoil; the angle of attack at cruise conditions; the twist of the wing, which is supposed to vary linearly spanwise; and the type of airfoil used in each wing segment. The type of airfoil refers to the standard NACA airfoils used for the base-of-reference model as well as reflexed airfoils, whose aft sections, originating from the standard ones, are deflected upwards by modifying the camber line of the standard airfoil, retaining the initial thickness distribution (Table 1).

Table 1. Design variables and bounding limits of the optimization process.

Design Variable	Bounding Limits
Static margin	10–30
Incidence angle of root airfoil (deg)	0–4
Overall Twist (deg)	2–5
Angle of attack in cruise conditions (deg)	0–4
Airfoil type at wing section 1	1–5 (increasing reflex from 1 to 5)
Airfoil type at wing section 2	1–5
Airfoil type at wing section 3	1–5
Airfoil type at wing section 4	1–5
Airfoil type at wing section 4	1–5

2.5. CFD Modelling

The performance of the optimized configuration, produced by the covariance matrix adaptation evolution strategy (CMA-ES), is validated through CFD modelling studies. The model is then considered the 'clean' configuration, which stands as a reference for integration with the nacelle as well as the nacelle/ducted fan propulsion system. CFD simulations are performed on the open-source OpenFOAM® platform [40]. Two computational approaches are adopted herein. Firstly, a steady-state solution is achieved using the SimpleFoam solver, which is based on the SIMPLE (semi-implicit method for pressure-linked equations) method. This is based on pseudo-time marching, thus achieving steady-state results in a short time. Therefore, it is preferred for analyzing steady-state flows such as the one around the clean configuration without the propulsion system. Conversely, a comprehensive performance study of the final configuration, i.e., the model with the integrated ducted fan standing in for the propulsion system, requires a temporally transient solver. For this purpose, the PimpleFoam solver is used to produce a steady-state flow solution by solving the transient Navier–Stokes equations with a high Courant number. The PIMPLE algorithm is a merge of SIMPLE (which is developed for steady physics) and PISO (which provides temporally accurate results). The main idea behind the PIMPLE loop is to find a fully converged steady-state solution with under-relaxation at each time step. Thus, the discretized momentum and continuity equations are converged at each time step, and the algorithm remains numerically stable. Nevertheless, by using large time steps which correspond to about 1 degree of rotation of the sliding mesh, we violate the CFL < 1 condition, which results in less accurate results in terms of the physics of the flow, which in general is temporally unsteady. It is a compromise in accuracy, which is common practice in many codes, so that we can estimate the performance of rotors in terms of average loading. The method is well customized for rotating machinery applications where a part of the computational mesh around the rotor, which is confined within cylindrical boundaries, slides over the stationary remaining mesh sharing the overlapping boundaries. Thus, equations are discretized and solved on a stationary frame of reference by simply taking into consideration the motion of the computational cells that are moving. The wall clock execution time is reduced by selecting a high time step (corresponding to about 1 degree of rotation). Thus, a transient simulation, which leads to statistically stationary results (after about 10–15 revolutions of the EDF rotor), is enabled through the use of a high-performance computing platform (ARIS).

3. Results and Discussion

3.1. Parametric Invsetigation

As already mentioned, the parametric investigation aims at the determination of the geometric characteristics which enable the operation of the UAV at flight speeds close to Mach number 0.1. In the preliminary analysis, the aspect ratio and the Oswald factor values are 2 and 0.8, respectively. The former is considered a typical value for BWB aircrafts, while the latter is set to a relatively conservative value regarding the expected lift distribution. For the wetted surface area of the UAV, a range between 8 and 18 m^2 is considered acceptable for the specific configuration. The Mach number varies between 0.05 and 0.35. For this range of flight speeds, the steps described in Section 2.1 are applied to derive the distributions of the ratios C_L/C_D and $C_L{}^{1/2}/C_D$. The positions of the maximum values in these plots are used for determining the velocities that maximize endurance and range, respectively. For the present analysis, the weight of the UAV is assumed to be 205 kg.

The results of the parametric analysis are presented in Figure 2. The plots illustrate the distributions of the ratios C_L/C_D and $C_L{}^{1/2}/C_D$ as a function of the Mach number for three different values of wetted surface area, namely 8, 12.5 and 18 m^2. These values correspond to the mean aerodynamic chords (MOCs) 2, 2.5 and 3 m, respectively. From these plots, it is shown that the intermediate value of wetted surface area leads to a UAV with a maximum endurance operating close to a Mach number equal to 0.1. The corresponding value of the lift-to-drag ratio is approximately 12. The value of the maximum ratio does not significantly change for different surface areas. On the other hand, although reducing the wetted surface area might be considered a requisite for the UAV, it results in narrower distributions, making the optimum Mach number somewhat more sensitive. Figure 2b indicates that if the final selection for the UAV's wetted surface area is close to 12.5 m^2, the range of the UAV is optimized at a Mach number close to 0.15.

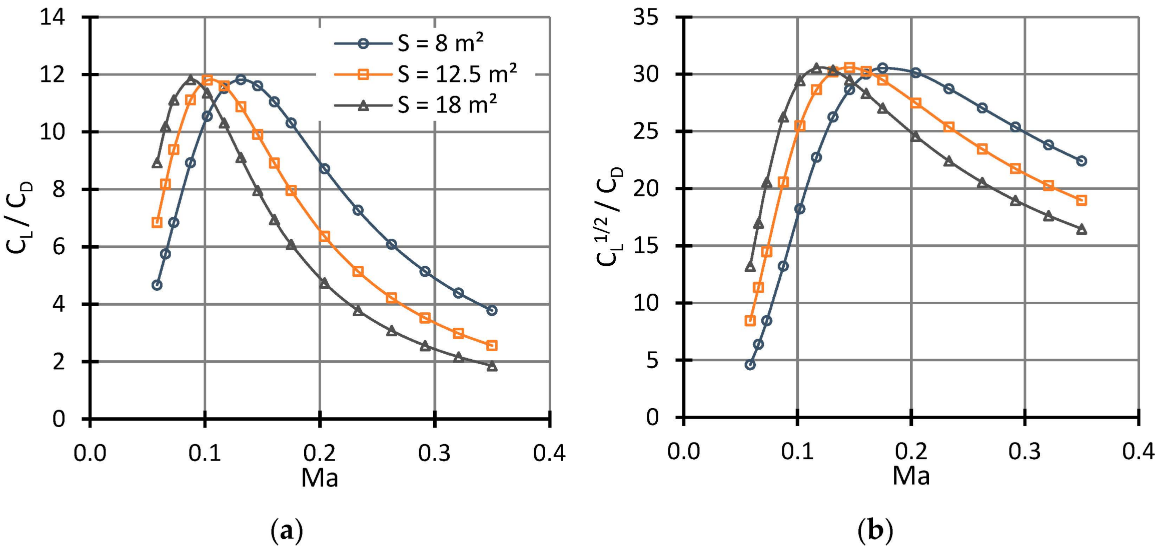


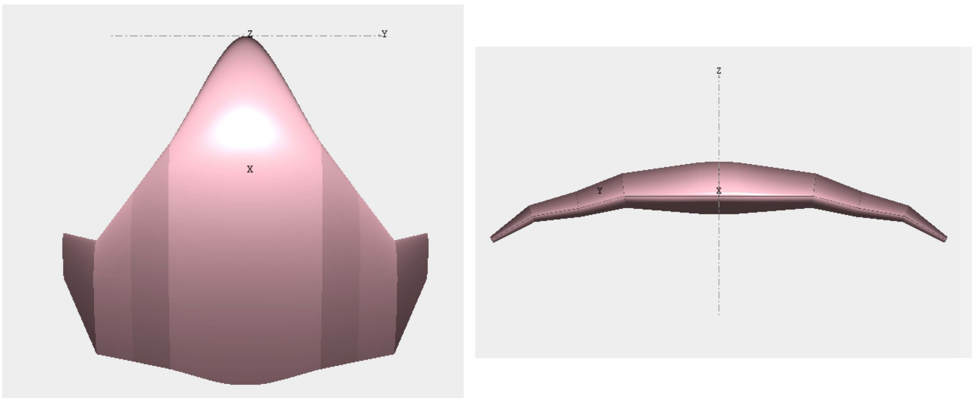
Figure 2. (**a**) C_L/C_D as a function of Mach number for S_{wet} = 8, 12.5 and 18 m^2; (**b**) $C_L{}^{1/2}/C_D$ as a function of Mach number for S_{wet} = 8, 12.5 and 18 m^2.

3.2. Base-of-Reference Xflr5 Model

A BWB UAV model, considered as the base-of-reference configuration, is built within the environment of the Xflr5 open software (version 6.60) [37]. The model consists of typical NACA airfoils, i.e., NACA2412 for the central part and NACA0012 for the outer ones. The dimensions and the geometrical properties of the airfoils used for the model are summarized in Table 2, while a sketch of the UAV is presented in Figure 3. The central part of the model IS refined using additional airfoils produced by applying a typical b-spline-based smoothing algorithm to the offset the airfoils in the spanwise direction. The anhedral considered in the outer part of the aircraft (Figure 3b) is used with the aim of reducing the induced drag and promoting stability in the spanwise direction. As regards airfoil selection, NACA 2412 is well suited to low-speed/light aircraft, as it produces the maximum lift-to-drag ratio at an angle of attack equal to 5 degrees, which is typical for the cruise flight of drones. NACA 0012, which is used for the outer part of the configuration, is primarily incorporated for stability and control purposes. In general, NACA four-digit airfoils are thick airfoils, advantageous at low speeds, with benign stall characteristics, and they are insensitive to non-smooth surfaces. With respect to stability issues, the center of pressure has limited movement over a wide range of angle of attack [41,42].

Table 2. Airfoils and their geometrical properties in the base-of-reference UAV model.

Section	NACA Airfoil	Semi-Span Position (m)	Chord Length (m)	Offset (m)	Dihedral (°)
1	2412	0.0	4.6	0.0	0.0
2	2412	0.2	4.5	0.1	0.1
3	2412	1.0	3.0	1.4	−5
4	2412	1.5	2.05	2.05	−10
5	2412	2.0	1.5	2.7	−30
6	0012	2.5	0.6	2.6	0



(**a**) (**b**)

Figure 3. Initial (base-of-reference) UAV configuration: (**a**) planform and (**b**) frontal area of the BWB UAV model (scaled dimensions).

The model is developed to match the wetted area needed for optimizing endurance at a Mach equal to 0.1. The wetted area is 13.11 m^2, while the aspect ratio and the mean aerodynamic chord are 1.87 and 3.198, respectively. For the above values, the series of calculations described in Section 2.2 are performed so that the analytical approach can be validated, at least in terms of lift-to-drag ratio. A direct comparison is presented in Figure 4b. Figure 4 also presents the drag, lift and momentum coefficients with the angle of attack as predicted by Xflr5. The zero-lift angle of attack has a negative value, while, due to the negative momentum of the zero angle of attack, the configuration needs to be trimmed.

For this purpose, the optimization algorithm described in Section 2.3 is incorporated, and the resulting geometry is examined utilizing computational fluid dynamics simulations.

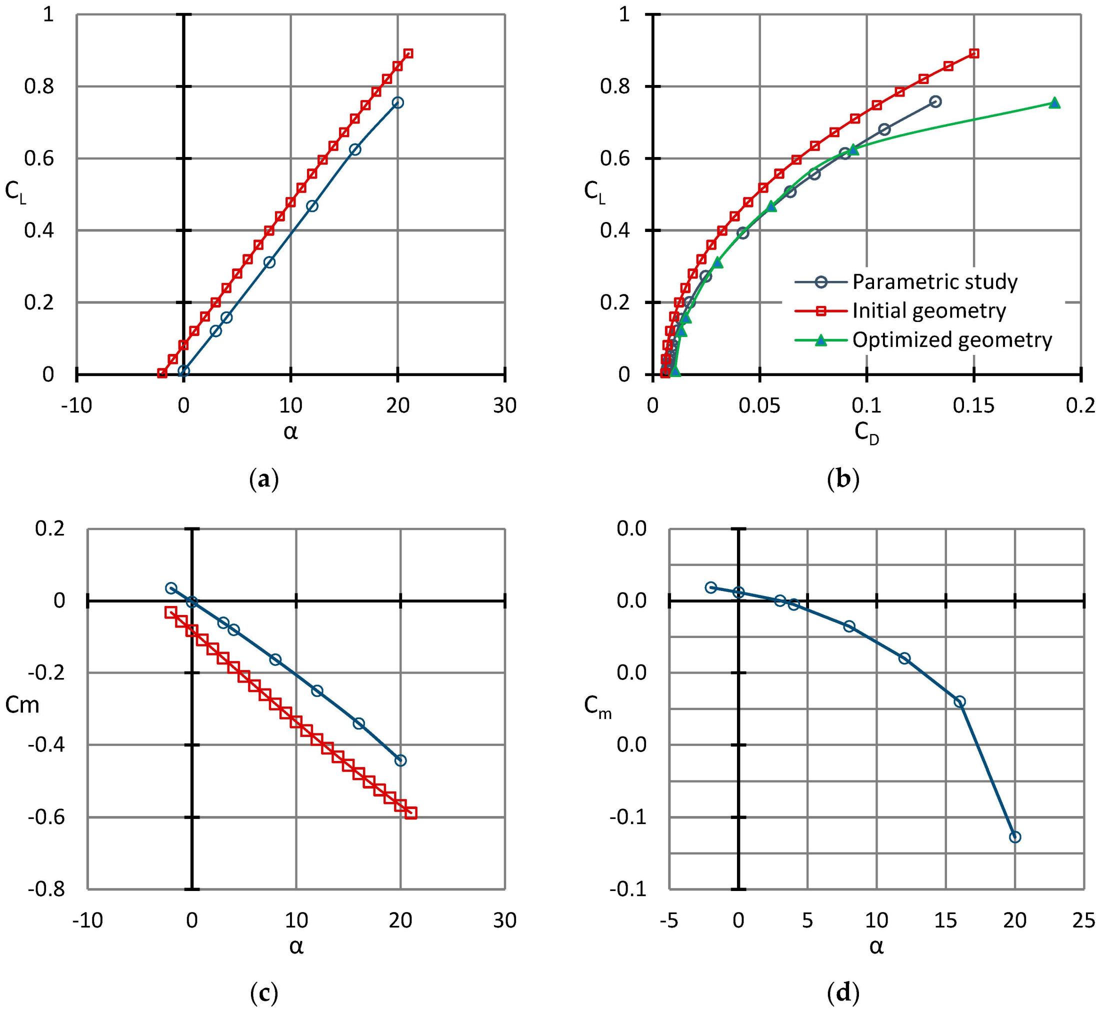


Figure 4. Aerodynamic coefficients and drag polars of the base-of-reference and the clean, optimized (trimmed) configuration: (**a**) C_L as a function of angle of attack; (**b**) C_L as a function of C_D; (**c**) C_m as a function of angle of attack; (**d**) CFD results of C_m as a function of angle of attack for the trimmed configuration only. In this latter case, momentum is measured from the center of gravity of the UAV.

3.3. Clean Configuration (C1): Optimization and CFD Results

For all the CFD models, numerical simulations are performed for a flight speed equal to 77.75 knots (IAS), which corresponds to the velocity that maximizes the endurance of the clean configuration, according to the analytical calculations in Section 2.1. CFD modelling was first applied to the 'clean' configuration (C1) produced by the optimization algorithm, as discussed in Section 2.4. For the present model, the cost function was minimized for an angle of attack, incidence angle and twist equal to 1.2, 1.52 and 2.44 degrees, respectively. The updated reflexed airfoils are presented in Figure 5. The final half of the UAV geometry is shown in Figure 6a.

The computational mesh is a mixed-type unstructured one around the whole aircraft, with about 30 inflation layers, on the order of 10 million cells. The first node lies in the laminar sublayer, namely y+ < 10, so that the low-Reynolds turbulence model, as proposed by Spalart & Almaras [43], is enabled. The specific turbulence model is one equation, a numerically stable model well customized for aerospace applications. The imposition of optimal values of boundary conditions for turbulence, far upstream of aircraft, are proposed in [44]. OpenFoam code is benchmarked against experimental results and the CFX-5 code for relevance, using both steady and unsteady swirling flows around water turbines in [45]. The simulation is temporally steady. The turbulent kinematic viscosity at the inlet is set at three times the molecular viscosity, i.e., $\mu_t = 4.5 \times 10^{-5}$ m^2/s. The performance diagrams showing the lift, the C_L, the drag, the C_D and the moment coefficient C_m as a function of the angle of attack, α, are depicted in Figure 4.

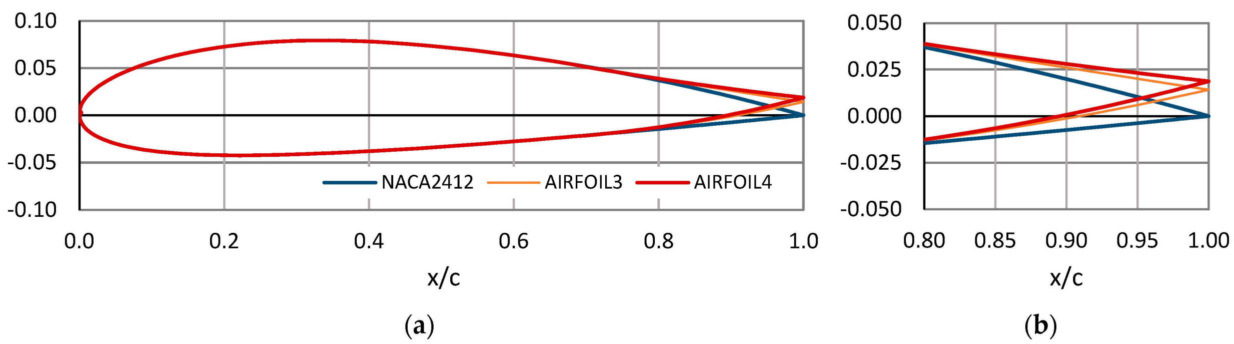


Figure 5. Standard and reflexed NACA2412 airfoils used for the aft region (airfoil 3 is used for a very small central region close to the axis of symmetry: (**a**) airfoils geometry; (**b**) reflexed region).

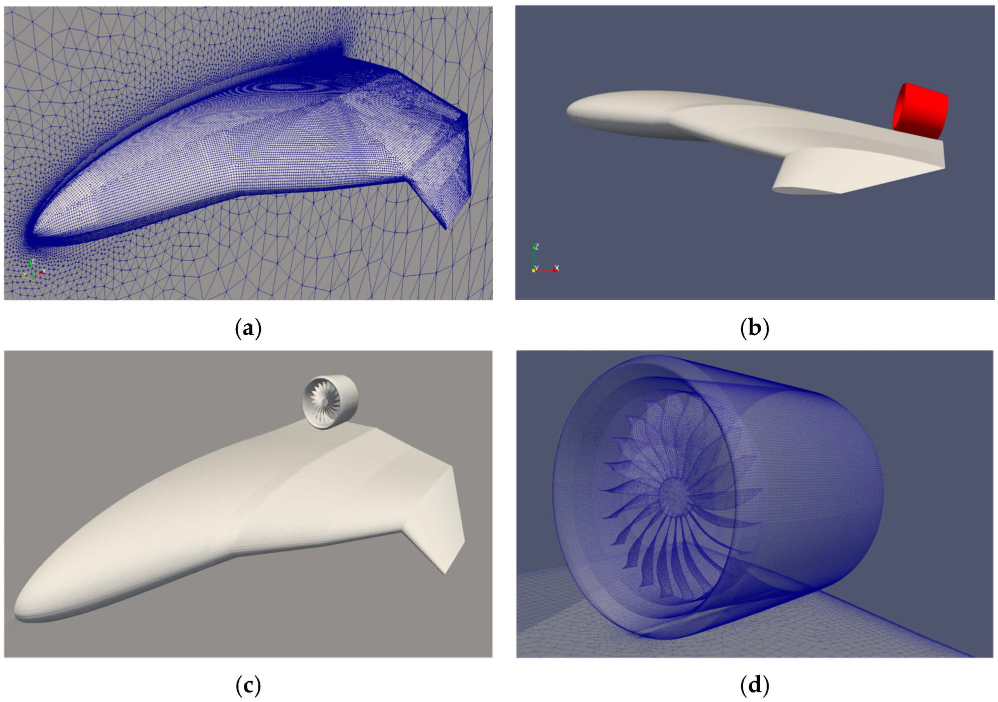


Figure 6. UAV configurations: (**a**) 'clean' UAV configuration (C1); (**b**) UAV with nacelle integration (C2); (**c**) UAV with integrated EDF, i.e., nacelle and rotor (C3); (**d**) ducted fan.

For the specific configuration, the center of gravity lies at x_{cg} = 2 m, measured from the nose of the aircraft, while the aerodynamic center lies at x_{ac} = 2.26 m. The non-dimensional static margin with respect to mean aerodynamic chord (MAC = 3.2) is 0.081. This value is marginally outside the typical range for static longitudinal stability that, according to [46], lies between 0.1 and 0.3. The rate of change of the pitching moment coefficient with respect to angle of attack is $C_{ma} = -0.086$, when a typical range is considered to be between -0.3 and -1.5 [46]. From this analysis, it is deduced that for the current configuration, despite being longitudinally statically stable, an augmentation of stability through either forward movement of the center of gravity or additional reflex of the airfoils is advisable.

The results shown in Figure 4 indicate that in this form, the UAV model is trimmed at a positive angle of attack equal to 3.9° and its stall angle is above 20° while it remains longitudinally statically stable, even at high α values, without the pitch-up tendency which is typical of BWB designs with back-swept wings [36].

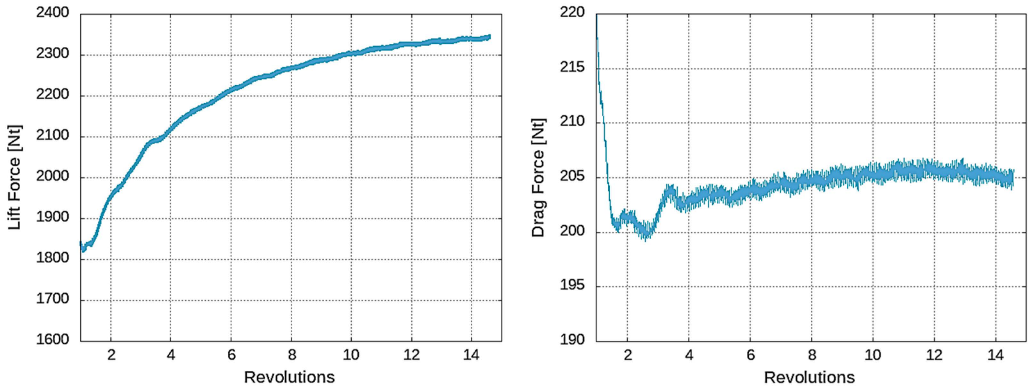
3.4. CFD Results of the UAV Model with Mounted Nacelle (C2) and Ducted Fans (C3)

The first of these models (C2) is presented in Figure 6b. It consists of the clean configuration and the nacelle of the EDF mounted on the upper side, close to the edge of the fuselage. In this way, it is thought that noise levels will be mitigated while the exhaust air from the ducted fan will contribute to the wake filling, thus reducing total drag. The computational mesh is a wall function type. Namely, the first node off the wall lies in the logarithmic part of the boundary layer for the sake of the reduction of required computational resources. This model also serves as a reference model with respect to the final configuration, where the EDFs will be integrated. For the current simulations, the angle of attack is set to 4°, a value which corresponds to near-level flight conditions. As can be seen in Figure 6a, the flow on the upper part of the nacelle is partially detached as well as the flow near the mounting point of the nacelle on the fuselage near the trailing edge.

Model case C3, shown in Figure 6c,d, stands for the CFD model where EDFs are integrated with the fuselage of the clean configuration. The ducted fan is supposed to be electrically driven; therefore, additional components will need to be integrated into future versions of this UAV model. In the current case, the propulsion system consists solely of the nacelle, formed by the revolution of a NACA0012 airfoil and a high-solidity 21-blade fan. This high-solidity rotor was selected for mitigation of uneven aerodynamic loading due to the development of the boundary layer on the fuselage. The casing's outer diameter is 0.5 m. Two ducted fans are considered to be mounted symmetrically on the upper part of the fuselage; however, to reduce the computational cost, only half of the geometry is modeled. For the present simulations, the angle of attack is set at 4°, while the rotational speed is set at N = 5500 rpm.

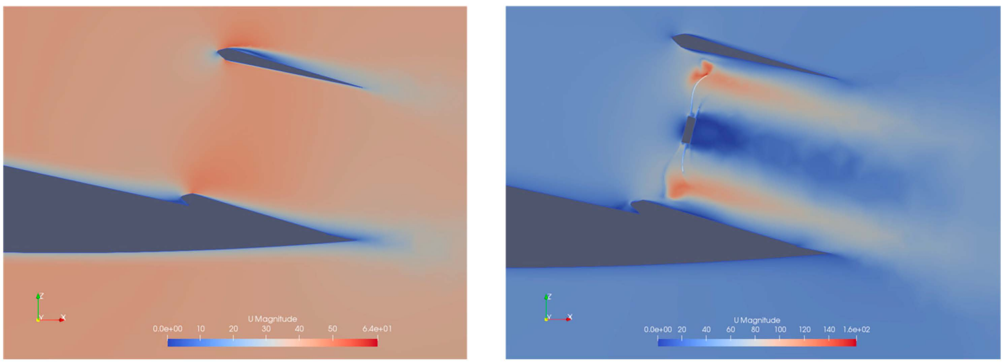
The computational mesh consists of an inner rotating part (Figure 6d) which slides over the remaining stationary part, used previously for the modelling of the C2 configuration. In this way, the governing equations are discretized on a stationary frame of reference by taking into account the rotation of control volumes for mass and momentum flux calculation. Thus, the formulation of momentum equations is simplified, since the addition of Coriolis and centrifugal force is not needed while the velocity vector is measured on the absolute frame of reference. As derived from Figure 7, the lift and drag forces on the fuselage are stabilized after 14 revolutions, which corresponds to a physical time of 0.1526 s. Since the free stream velocity is set to 40 m/s, the flow is estimated to run about 2 length units of the whole model in this time interval, which is considered adequate for the forces to reach a plateau.

Compared to the configuration without the fan (Figure 8a), it is readily concluded that the local flow acceleration created by the rotor contributes to the attachment of the flow on the upper part of nacelle and the jet stream at the fan exit to the reduction of wake mixing. This results in higher lift production for the C3 over the C2, while the drag on the nacelle for C2 turns into a slight thrust for C3 due to the jet effect. These findings are clearly depicted in Tables 3 and 4. It is also worth mentioning that the simulation for the C3 design corresponds to a sustained unaccelerated 1.29 g turn, since there is almost a force balance among the fuselage/nacelle/rotor in trajectory/drag direction (Table 4). These flight conditions are considered realistic, since there is a load balance between the UAV components. In terms of lift-to-drag ratio, it is verified that a considerable increase is achieved in the final configuration (C3) over the previous ones (C1 and C2), which can be attributed to the diversion of the flow inboard and to the acceleration of the boundary layer. The lift-to-drag ratios calculated by the results presented in Tables 3 and 4 for the C1, C2 and C3 configurations are 10.33, 9.64 and 12.92, respectively. Thus, the flow encounters thicker airfoils at higher speeds that result in higher aerodynamic loading. The redirection of the flow is verified by the increase of the lateral velocity component for the C3 over the C2 configuration, which is revealed in Figure 9. The higher loading of the fuselage for the C3 over C2 model is also verified by the higher suction on the upper fuselage, shown in Figure 9.



(**a**) (**b**)

Figure 7. Forces calculations with the number of revolutions: (**a**) Lift force; (**b**) Drag force.



(**a**) (**b**)

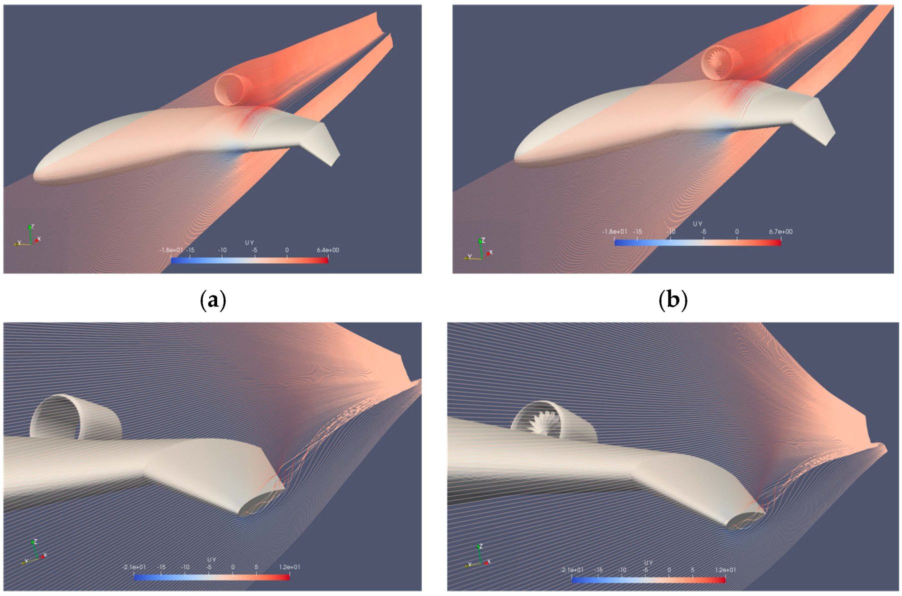
Figure 8. Contours of velocity magnitude at the vicinity of the nacelle and the ducted fan: (**a**) nacelle integration (C2); (**b**) ducted-fan integration C32 (Flight conditions IAS = 77.75 knots, $\alpha = 4°$).

Table 3. Lift force [N] on each component for several configurations.

Configuration	Fuselage	Nacelle	Skin and Nacelle	Fan	Total
C1	2046	-	2046	-	2046
C2	2000	150	2150	-	−2150
C3	2346	177	2523	75	2598

Table 4. Drag force [N] on each component for several configurations.

Configuration	Fuselage	Nacelle	Skin and Nacelle	Fan	Total
C1	198	-	198	-	198
C2	175	48	223	-	223
C3	205	−4	201	−214	−13



(**a**) (**b**)

(**c**) (**d**)

Figure 9. Streamlines around the UAV: (**a**) At the mid-semi-span of C2 (y = −1.2 m, maximum lateral velocity U_y = 6.4 m/s); (**b**) At the mid-semi-span of C3 (y = −1.2 m, maximum lateral velocity U_y = 6.7 m/s); (**c**) Close to the wingtip of C2 (y = −2.3 m, maximum lateral velocity U_y = 12 m/s); (**d**) Close to the wingtip of C3 (y = −2.3 m, maximum lateral velocity U_y = 12 m/s).

The distribution of the pressure coefficient (Cp) is depicted at several spanwise positions in Figure 10. A strong adverse pressure gradient, which decelerates the boundary layer on the both pressure side and the suction side from the symmetry plane to at least 50% of the span is observed. At about 75% of the chord, the Cp on the suction side crosses the Cp on the pressure side and increases up to the trailing edge. Thus, the loading of the airfoil decreases. This trend is alleviated in the C3 case where the suction on the upper side is retained due to the acceleration of the boundary layer by the BLI effect. In front of the Cp cross point, the C3 case produces higher loading, since the pressure distribution on the pressure side is higher and on the suction side lower than the respective sides in comparison to the C1 and C2 cases. At 95% of the span, there is no evident difference in pressure distribution between the C1, C2 and C3 cases. It is worth noting that the flow remains attached to a weak adverse pressure gradient, which hints at a low-intensity wingtip vortex and reduced spanwise flow at the wingtips.

Finally, a preliminary assessment of the effect of the boundary layer ingestion on the performance of the EDF is attempted through monitoring the forces and moments exerted on the rotor in the case of the isolated and mounted EDF on the aircraft (Table 5). The moment about the z-axis (pitch up) by the mounted EDF verifies the uneven loading of the rotor due to the distorted velocity profiles it encounters, which is not observed in the isolated EDF case, while there is also a reduced torque moment in the mounted EDF case in comparison to the isolated one, which hints at an increase in EDF performance due to BLI.

Table 5. Forces and moments about the rotor axis of the EDF at cruise conditions.

Configuration	Force (x-Axis) (N)	Force (y-Axis) (N)	Force (z-Axis) (N)	Moment (x-Axis) (Nm)	Moment (y-Axis) (Nm)	Moment (z-Axis) (Nm)
Mounted EDF	−107.19	4.24	37	45.22	0.24	−71.56
Isolated	−107.59	−2.19	15.5	68.47	−0.92	−4.54

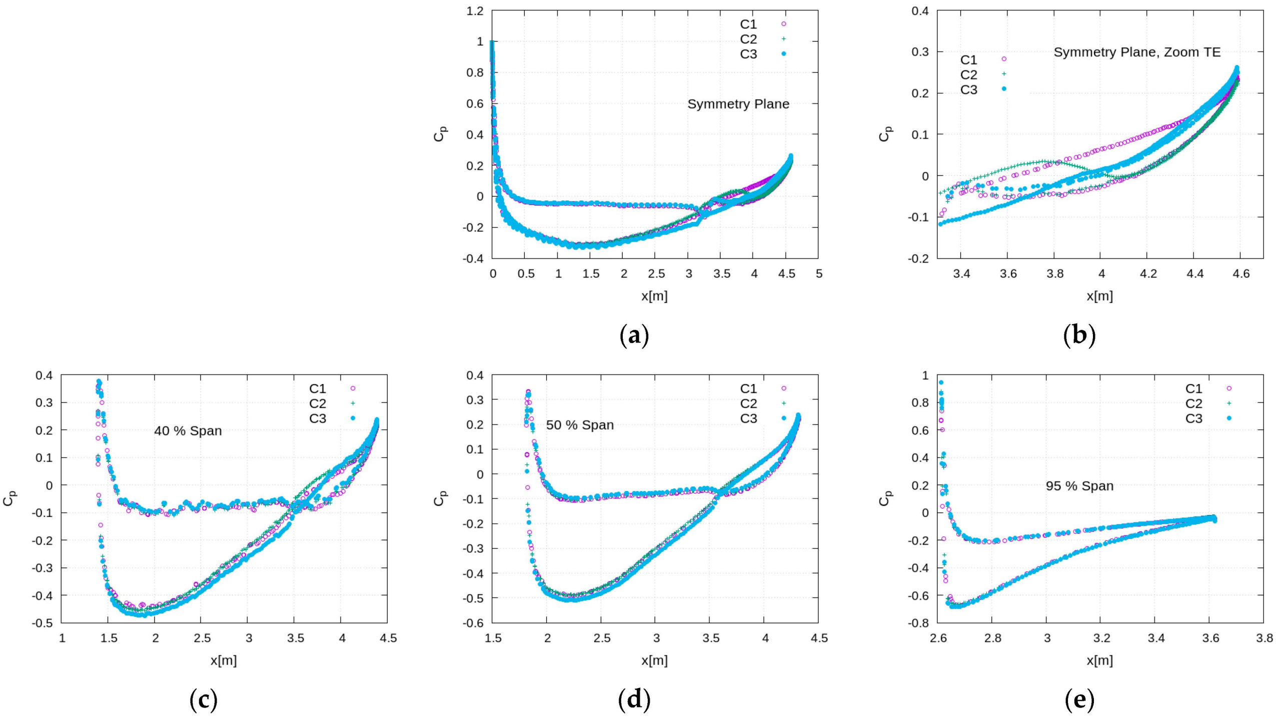


Figure 10. Pressure coefficient distribution at several spanwise positions: (**a**) symmetry plane, (**b**) symmetry plane (focused), (**c**) 40% of the span, (**d**) 50% of the span and (**e**) 95% of the span.

4. Concluding Remarks and Future Work

The present work deals with aerodynamic design aspects of a BWB UAV. These results aim to contribute to the comprehensive understanding of the aerodynamic behavior of the UAV, offering insight into its potential performance in real-world applications.

As a first step, a parametric investigation is carried out to determine the wetted area necessary for the UAV to achieve maximum endurance at a Mach number close to 0.1. These preliminary results indicate that for a typical aspect ratio considered for BWB configurations around 2, the necessary wetted area is about 12.5 m^2. Based on this finding, as well as on design considerations related to the aerodynamic efficiency and stability of BWB aircrafts, a base-of-reference configuration is drawn. The configuration consists of a combination of NACA airfoils. Its performance is examined by incorporating a typical panel method. Both development and analysis are conducted by utilizing the environment and the solvers available in Xflr5. The produced UAV needs to be trimmed for positive momentum at zero angle of attack. For this purpose, an optimization algorithm is employed to minimize the value of a cost function, which takes into account the absolute difference between maximum takeoff weight and lift, the drag-to-lift ratio and the momentum coefficient measured around the center of gravity. Design variables are the incidence angle of the root airfoil, the angle of attack at cruise conditions, the twist of the wing, which is supposed to vary linearly spanwise and the type of airfoil used in each wing's segment. The produced 'clean' configuration is characterized by an angle of attack, an incidence angle and twist equal to 1.2, 1.52 and 2.44 degrees, respectively. In addition, customized reflexed airfoils are produced to form a negative pitching moment, contributing in this way to the longitudinal stability of the UAV. To verify improvements, the polar diagrams of the clean configuration are estimated by CFD modelling. Nacelle and EDF integration are considered then. Simulations are performed by incorporating an incompressible flow solver available in the OpenFoam platform. When the EDF integration is modeled, the static mesh used for simulating the flow past the whole geometry is coupled with a sliding, moving mesh used for capturing fan rotation. Lift and drag forces are compared to study the effects of the integration of the propulsion system. Forces are analyzed to those produced by the fuselage, the nacelle and the fan. Results demonstrate that due to the jet flow behind the duct and the entrainment of ambient air, the lift-to-drag ratio of the final configuration increases substantially.

The design studies presented herein offer the opportunity to evaluate the results of the adoption of a fully electric propulsion system in terms of aerodynamic characteristics, relating the lift and drag forces as well as the flow field characteristics in the vicinity of the EDF. As regards the clean configuration, its polar diagrams indicate that it remains longitudinally stable at angles of attack close to 20°. This can be attributed to the high anhedral angle of the outboard section of the polyhedral wing which prevents stalling at high angles of attack. When the EDF is mounted on the UAV's fuselage, the spanwise flow which potentially leads to pitch-braking and loss of stability is substantially mitigated, since the flow is accelerated by the EDF, which diverts the flow inboard. The wingtip vortex, which is also fed by the spanwise flow, is also weakened. According to the present findings, the direct mounting of the EDF on the UAV's fuselage is conducive to higher loading at a minimal drag penalty, and thus a considerable increase of the lift-to-drag ratio is achieved. Finally, by evaluating the forces and moments about the rotor axis of the EDF and comparing the rotor pressure distribution under isolated and cruise conditions, an indication of the BLI mechanism on the propulsion system can be observed.

In this study, the aerodynamic characteristics of a BWB UAV are presented, providing information on the integration of a ducted fan-based propulsion system with the aircraft's fuselage towards the proof-of-concept of a specific configuration. However, certain aspects require further investigation to achieve a more comprehensive understanding. In future work, a more detailed analysis will be carried out, aiming to further explore the benefits of the BLI concept in terms of both aircraft aerodynamics as well as of the propulsion system performance.

Author Contributions: Conceptualization, methodology, writing—review and editing, N.L. and A.V.; conceptualization, methodology and supervision, I.T. and T.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are unavailable due to privacy and ethical restrictions.

Acknowledgments: This work was supported by computational time granted by the National Infrastructures for Research and Technology S.A. (GRNET S.A.) in the National HPC facility—ARIS—under project ID 1111.

Conflicts of Interest: The authors declare no conflicts of interest.

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